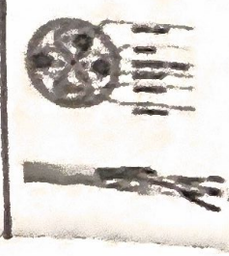
Coaxial cable	RG-6 RG-59 RG-11	10-100Mbps	• High bandwidth • Immune to interference • Low loss bandwidth • Versatile **Disadvantages** • Limited distances • Cost • Size is bulky	Speed of signal is 500m Television network High speed internet connections	
fibre optics cable	Single mode Multi mode	100Gbps	**Advantages** • High speed • High bandwidth • High security • Long distance **Disadvantages** • Expensive • Requires skilled installers	• Maximum distance of fibre optics cable is around 100meters	

Cable type	Category	Maximum Data Transmission	Advantages/ Disadvantages	Application/Use	Image
UTP	Category 3	10 bps	**Advantages** • Cheaper in cost • Easy to install as they have a smaller overall diameter **Disadvantages** • More prone to (EMI) Electromagnetic interference and noise	10Base-T Ethernet Fast Ethernet, Gigabit Ethernet Fast Ethernet, Gigabit Ethernet	
	Category 5	Up to 100 Mbps			
	Category 5e	1Gbps			
STP	Category6/6a	10Gbps	**Advantages** • Shielded. • Faster than UTP. • Less susceptible to noise and interference **Disadvantages** • Expensive • Greater installation effort	Gigabit Ethernet, 10G Ethernet (55m) Widely used in data centres	
SSTP	Category 7	10Gbps		Gigabit Ethernet, 10G Ethernet (100m)	

Exercise: 03
Date: 30/07/26

Types of network cables

AIM: Study of different types of Network cables.

(a) understand different types of network cables

Different type of cables used in Networking are

1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted pair (STP) cable
3. coaxial cable
4. Fibre optic cable.

(b) Make your own Ethernet cross-over cable/ straight cable.

Tools and parts needed:

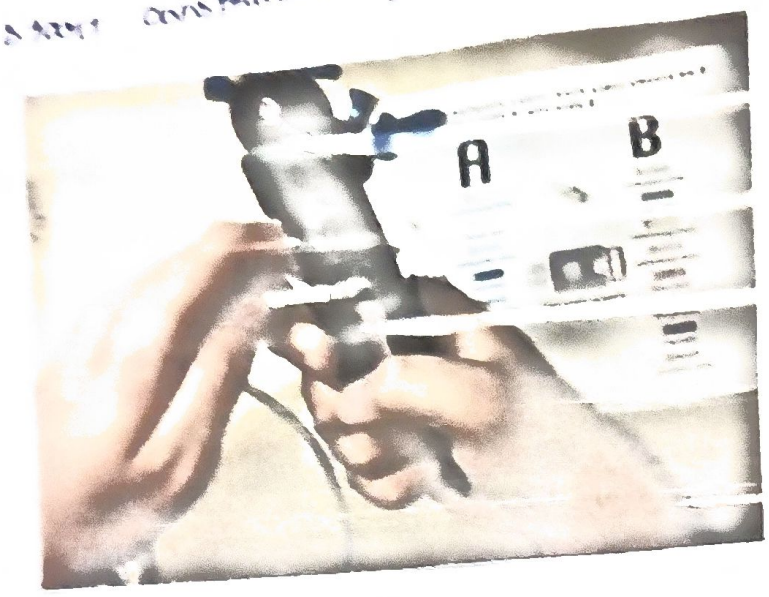
- Ethernet cabling - CAT 5e is certified for gigabit support, but CAT 5 cabling works as well, Just over shorter distances.
- Two RJ45 plugs.
- Optional two plug shields.

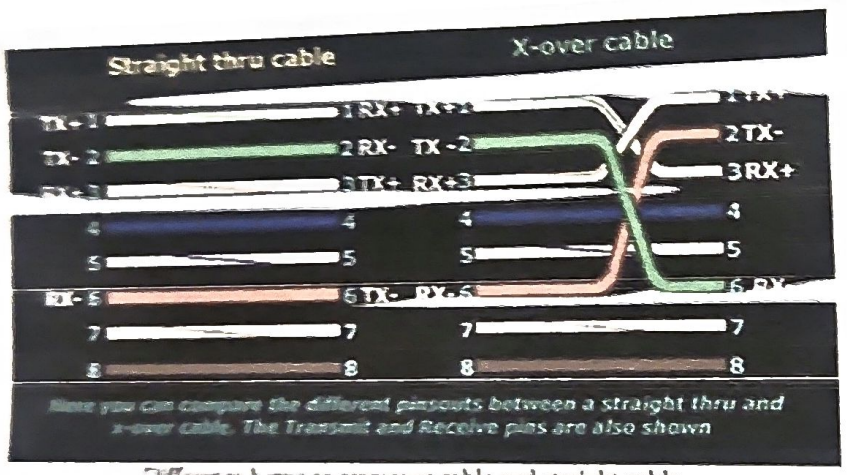
Procedure

Step: 1 To start construction of the device, bigin by threading shields onto the cable.

Step: 2: Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

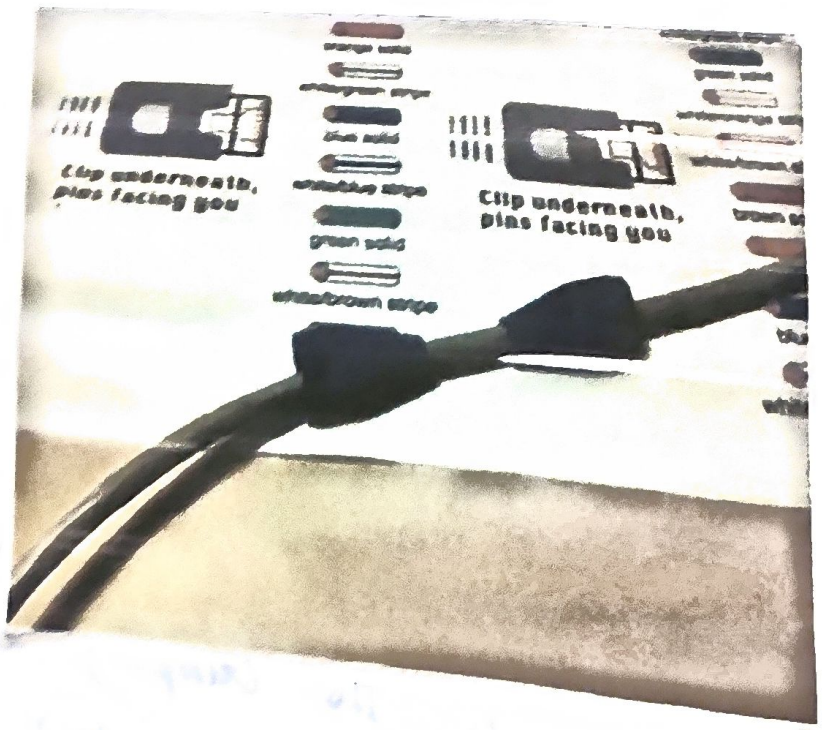
For a step by step construction of the above





Difference between crossover cable and straight cable

Next strip approximately 15 cm of cable shielding both ends.



Step: 3 After you
these should
to the sheet
One end sho

Step 4. Once
together in a
stick out
to create an
Placing these
mossing up

Step: 5: nex
notch at
just over
that mean
shielding

Step 6: Af
inside the
to tool push

Step 7: Last
diagram B
diagram A

Step 3 = Right you will need to untangle the wires. there should be four twisted pairs separating back to the sheet, arrange them from top to bottom one end should be in arrangement A and the other in

Step 4. Once the order is correct, bunch them together in a line, and if there are any that stick out further than others. Snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order

Step: 5: next push the cable right in to the notch at the end of the plug needs to be just over the cable shielding and if it isn't that means that you stripped off too much shielding

Step 6: After the wires are securely sitting inside the plug insert it into the crimping tool push down

Step 7: Lastly, repeat for the other end using diagram B (to make a crossover cables) / using diagram A (to make a straight through cable.

INDE

DATE	TEACHER'S / REMA
10/9/26	
19/26	
9/28	

Result

Thus a simple robot was designed and stimulated using circo packet tracer and the forwarding behaviour of hub and switch was analysed successfully